STUDENT REPORT

DETAILS

Name

THANGADIPALLY ANVITHA

Roll Number

20R01A6753

EXPERIMENT

Title

WEEK 6

Description

6. Demonstration of clustering rule process on dataset iris.arff using simple k-means.

Aim: This experiment illustrates the use of simple k-mean clustering with Weka explorer. The sample data set used for this example is based on the iris data available in ARFF format. This document assumes that appropriate preprocessing has been performed. This iris dataset includes 150 instances.
Steps involved in this Experiment
Step 1: Run the Weka explorer and load the data file iris.arff in preprocessing interface.
Step 2: Inorder to perform clustering select the 'cluster' tab in the explorer and click on the choose button. This step results in a dropdown list of available clustering algorithms.
Step 3 : In this case we select 'simple k-means'.
Step 4: Next click in text button to the right of the choose button to get popup window shown in the screenshots. In this window we enter six on the number of clusters and we leave the value of the seed on as it is. The seed value is used in generating a random number which is used for making the internal assignments of instances of clusters.
Step 5 : Once of the option have been specified. We run the clustering algorithm there we must make sure that they are in the 'cluster mode' panel. The use of training set option is selected and then we click 'start' button. This process and resulting window are shown in the following screenshots.
Step 6 : The result window shows the centroid of each cluster as well as statistics on the number and the percent of instances assigned to different clusters. Here clusters centroid are means vectors for each clusters. This clusters can be used to characterized the cluster.For eg, the centroid of cluster1 shows the class iris.versicolor mean value of the sepal length is 5.4706, sepal width 2.4765, petal width 1.1294, petal length 3.7941.
Step 7: Another way of understanding characterstics of each cluster through visualization ,we can do this, try right clicking the result set on the result. List panel and selecting the visualize cluster assignments.

Description:
Clustering is a process of partitioning a set of data (or objects) into a set of meaningful sub-classes, called clusters.
K-Means clustering intends to partition *n* objects into *k* clusters in which each object belongs to the cluster with the nearest mean.

Procedure:

Step 1: Load appropriate dataset into WEKA

Step 2: Go to Cluster tab→choose→SimpleKmeans

Procedure and Overvation:

```
step1: Take the mean value

step2: Find the nearest number of mean and put in cluster

step3: Repeat 1 and 2 until we get the same cluster.

@RELATION iris

@ATTRIBUTE sepallength REAL


@ATTRIBUTE sepalwidth REAL

@ATTRIBUTE petallength REAL

@ATTRIBUTE petalwidth REAL

@ATTRIBUTE class {Iris-setosa,Iris-versicolor,Iris-virginica}

@DATA
5.1,3.5,1.4,0.2,Iris-setosa

4.9,3.0,1.4,0.2,Iris-setosa

4.7,3.2,1.3,0.2,Iris-setosa

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6.3,3.4,5.6,2.4,Iris-virginica

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6.7,3.0,5.2,2.3,Iris-virginica

6.3,2.5,5.0,1.9,Iris-virginica

6.5,3.0,5.2,2.0,Iris-virginica

6.2,3.4,5.4,2.3,Iris-virginica

5.9,3.0,5.1,1.8,Iris-virginica

%

%

%
```

Choose **SimpleKMeans** -init 0 -max-candidates 100 -periodic-pruning 10000 -min-density 2.0 -t1 -1.25 -t2 -1.0 -N 3 -A "weka.core.EuclideanDistance -R first-las

Cluster mode

- (●) Use training set
- ( ) Supplied test set — Set...
- ( ) Percentage split — % 66
- ( ) Classes to clusters evaluation
  - (Nom) class
- [x] Store clusters for visualization

Ignore attributes

Start | Stop

Result list (right-click for options)

- 11:49:35 - EM
- 11:50:19 - SimpleKMeans

Clusterer output

```
                petallength
                petalwidth
                class
Test mode:      evaluate on training data


=== Clustering model (full training set) ===


kMeans
======

Number of iterations: 3
Within cluster sum of squared errors: 7.817456892309574

Initial starting points (random):

Cluster 0: 6.1,2.9,4.7,1.4,Iris-versicolor
Cluster 1: 6.2,2.9,4.3,1.3,Iris-versicolor
Cluster 2: 6.9,3.1,5.1,2.3,Iris-virginica

Missing values globally replaced with mean/mode

Final cluster centroids:
                                        Cluster#
Attribute                 Full Data            0              1
                            (150.0)       (50.0)         (50.0)        (50.
==============================================================================
sepallength                 5.8433        5.936          5.006          6.5
sepalwidth                   3.054         2.77          3.418          2.9
petallength                 3.7587         4.26          1.464          5.5
petalwidth                  1.1987        1.326          0.244          2.0
class                  Iris-setosa Iris-versicolor     Iris-setosa  Iris-virgini




Time taken to build model (full training data) : 0 seconds

=== Model and evaluation on training set ===

Clustered Instances

0       50 ( 33%)
1       50 ( 33%)
2       50 ( 33%)
```

Open file... | Open URL... | Open DB...

Filter

Choose **None**

Current relation

Relation: iris
Instances: 150

Attributes:
Sum of weights:

Attributes

All | None | Invert | Pattern

No.		Name
1	[ ]	sepallength
2	[ ]	sepalwidth
3	[ ]	petallength
4	[ ]	petalwidth
5	[ ]	class

VI

Q) Who is responsible for running queries and reports against data warehouse tables?

End-users

Q) From where are classification rules extracted?

Decision tree

Q) Identify the method for incremental conceptual clustering.

COBWEB

Q) What is the full form of KDD?

Knowledge discovery database

Q) What is the process of removing the deficiencies and loopholes in the data called?

Extraction of data